VII. (points) Nuclear electric force makes an electron circle in classical hydrogen. Classical EM theory says accelerated electrons emit waves; the radiated power is

given by Larmor's formula ; is acceleration, vacuum light speed,

; the electron charge magnitude is

Neglect relativity and estimate a classical atom's lifetime . (Measured binding energy is , and electron rest mass is

Radiation from accelerated charges has important uses. Synchrotron radiation, widely used in modern research, comes from near-light-speed electrons curving in magnetic fields. Its core is the electron storage ring; bunches traverse bending magnets and other devices to make high-quality synchrotron light. The recently built Shanghai facility is a third-generation source, with electron energy and bunch current , circumference , cells (bending-magnet turns) , and dipole field . Find mean total radiated power per outlet

(Near-light-speed electrons obey Newton's second law if relativistic mass is constant; the Larmor formula becomes , where is total energy, and is electron rest energy.)

Near-light-speed ring electrons emit synchrotron radiation in a cone along the orbit tangent, with half-angle . Thus higher electron energy gives better directionality. For this facility, find the duration of a single electron's radiation received at a point along the radiation direction:

(Give all results in this problem to three significant figures.)